MSG No. 89.92 P_c422 [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `422`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,7,8,9,10,11,14]`

Table 3: Wyckoff site: `2c`, site symmetry: `422`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `2d`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[4,5,7,8,9,10,11,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,5,6]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[10,11,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,6,12,13]`
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,3,15,16]`
3	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[4,5,9,14]`
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[7,8,10,11]`

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6]`
2	$[0,\ 0,\ -z]$	`[4,5,7,8]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,10,11,14]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[12,13,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,3,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13,15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8]`
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,14]`
6	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[10,11]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[12,13]`
8	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[15,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$[-x,\ x,\ 0]$	`[2,5]`
3	$[x,\ -x,\ 0]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`
5	$\left[x,\ x,\ \frac{1}{2}\right]$	`[9,15]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[10,13]`
7	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[11,12]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[14,16]`

Table 11: Wyckoff site: `8k`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,15]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,13]`

continued ...

Table 11

No.	position	mapping
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	[3,12]
4	$\left[x,\ -x,\ \frac{3}{4}\right]$	[4,11]
5	$\left[-x,\ x,\ \frac{3}{4}\right]$	[5,10]
6	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,16]
7	$\left[x,\ x,\ \frac{3}{4}\right]$	[7,9]
8	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4]
2	$[0,\ x,\ 0]$	[2,7]
3	$[0,\ -x,\ 0]$	[3,8]
4	$[-x,\ 0,\ 0]$	[5,6]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[9,12]
6	$\left[0,\ x,\ \frac{1}{2}\right]$	[10,15]
7	$\left[0,\ -x,\ \frac{1}{2}\right]$	[11,16]
8	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[13,14]

Table 13: Wyckoff site: 8m, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[1,12]
2	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[2,15]
3	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,16]
4	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[4,9]
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,14]
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[6,13]
7	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,10]
8	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[8,11]

Table 14: Wyckoff site: 8n, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,12]
2	$\left[0,\ x,\ \frac{1}{4}\right]$	[2,15]
3	$\left[0,\ -x,\ \frac{1}{4}\right]$	[3,16]
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	[4,9]
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[5,14]
6	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[6,13]
7	$\left[0,\ x,\ \frac{3}{4}\right]$	[7,10]
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	[8,11]

Table 15: Wyckoff site: 8o, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,4]
2	$\left[\frac{1}{2},\ x,\ 0\right]$	[2,7]
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	[3,8]
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,6]
5	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,12]
6	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[10,15]
7	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[11,16]
8	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14]

Table 16: Wyckoff site: 16p, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[x,\ -y,\ -z]$	[4]
5	$[-x,\ y,\ -z]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[9]
10	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[10]
11	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[12]
13	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[15]
16	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[16]